

We explain the modifications from DSMC to DSBGK using the description of [Jun Li, *Fuel*, **250**: 154-159, 2019]. Please keep in mind that each molecule has $\vec{x}_i, \vec{c}_i, F_i, N_i$ and each cell has $n_k, \vec{u}_k, T_k$ (omit the subscript 'tr' here), and then try to understand when and how to update each of them:

1. **Initialization:** in DSMC we already have the initial $\vec{x}_i, \vec{c}_i$ for the position and velocity of molecule i , DSBGK additionally assigns F_i, N_i as follows. As we know which cell each molecule is located in according to $\vec{x}_i$ (assume in the cell k), and know the initial macroscopic properties $n_k, \vec{u}_k, T_k$ of each cell k , we have the equilibrium f^{eq} for the cell k , which, together with $\vec{c}_i$, determines $F_i = f^{eq}(\vec{c}_i, n_k, \vec{u}_k, T_k)$. Now, assign N_i as a constant N_{init} for all molecules and its value could be the same as in DSMC (larger N_{init} means less 'simulated' molecules in each cell on average). The initial values for $n_k, \vec{u}_k, T_k$ are usually uniform.
2. **Movement and collision TOGETHER during each Δt :** first update $\vec{x}_i$ according to a uniform motion in a straight line, which should be done differently from DSMC. We need to check whether during each Δt the molecule will cross the interfaces of its current cell; basically we need a subroutine to let the molecule move via several substeps (not just one step as in DSMC), each of which will move the molecule within a single cell to a position at the cell interface, followed by the next substep that moves it within the next adjacent cell; the starting position and ending position of the whole Δt are usually located inside cells (not at interfaces) and it might have only one substep if the molecule moves slowly or if we used a very small timestep Δt compared to the ratio of the cell size to the mean molecular speed. The point to divide Δt into several substeps is to capitalize the conservation laws to *adjust* the cell variables (but DSMC uses the summations of transient molecular variables to *calculate* the cell variables), which needs to use the increments (instead of transient values) of all molecules due to the intermolecular collisions occurred within the SAME cell and SAME Δt (of course any substep is within the same Δt). Such a subroutine of splitting Δt into several substeps will return $\Delta_k t_i$ for each substep (see the left figure), which is the fraction of Δt used by the molecule i to move inside the cell k ; we then can compute F_i^{new} after each substep according to Eq. (7) and compute N_i^{new} according to Eq. (8) and now can determine $\Delta_k N_i$, which is the weight increment of molecule i due to collisions inside cell k . Repeat the above updating of molecular variables $\vec{x}_i$ due to movement and F_i, N_i due to collisions for all molecules. Now, we are able to update the cell variables $n_k, \vec{u}_k, T_k$ using $\Delta_k N_i$ of all molecules, which ever moved in the targeted cell k during the current Δt , in a 'low-noise' algorithm of Eq. (9) according to the auto-regulation and conservation principles, where $\sum_{i \in k} \Delta_k N_i, \sum_{i \in k} \Delta_k N_i \vec{c}_i, \sum_{i \in k} \Delta_k N_i \vec{c}_i^2$ can be saved as auxiliary cell variables and accumulated within a single Δt while moving each molecule i . People might wonder why $\vec{c}_i$ is unchanged so far, as in LBM. It is unchanged until the molecule i reflects on the wall boundaries (see #3 below), as in DSMC not LBM (DSBGK has DSMC-LBM duality). Note that DSMC changes $\vec{c}_i$ but DSBGK changes N_i to reflect the intermolecular collision effect, for which please see the discussion of extrapolation of the acceptance-rejection scheme in [27th RGD Conf. Proc. 1333, 283-288 (2011)].
3. **Wall boundaries:** without loss of generality, we assume the wall boundaries are located regularly as the cell interfaces (e.g., divide a regular computational domain and wall boundaries using uniform $\Delta x, \Delta y, \Delta z$); the above subroutine should also identify when a molecule has a substep ending at a cell-solid interface (instead of a cell-cell interface). This will activate a subroutine for the boundary condition, where the reflection position $\vec{x}_i$ is known as the ending position of the corresponding substep and thus the properties (surface normal, temperature, moving velocity, etc) of the wall surface can be determined by the reflection position. We follow the DSMC algorithm to compute $\vec{c}_r$ and update $\vec{c}_i$ to $\vec{c}_r + \vec{u}_{wall}$ ($\vec{x}_i$ is unchanged during reflection and N_i is also unchanged to conserve the mass). Additionally, F_i should be updated after reflection. For this, we use the kinetic boundary condition of completely diffuse reflection as an example, where the distribution function is $f(\vec{c}_r) = f^{eq}(\vec{c}_r, n_{eff}, T_{wall})$, where n_{eff} is the effective wall number density and can be determined according to the mass conservation law for the reflection process using Eqs. (11) and (13). This n_{eff} can be saved for each cell's solid interface and used for all molecules reflecting on the same solid interface during the same Δt , but will change for the next Δt and of course changes with the specific solid interface in general. We predetermine n_{eff} for all solid interfaces at the beginning of each Δt and use it in $f^{eq}(\vec{c}_r, n_{eff}, T_{wall})$ to determine $F_i^{new} = f^{eq}$ after molecular reflection.
4. **Open boundaries:** molecules will be removed as in the DSMC when they cross the open boundaries during each Δt . At the end of each Δt , new molecules will be added and assigned with $\vec{x}_i, \vec{c}_i$ as in DSMC. We also need to assign F_i as in the initialization according to the local f^{eq} (e.g., might have different n_{local} for different open locations). Additionally, we can simply let $N_i = N_{init}$, or use different N_{const} for different open locations such that $N_i/n_{local} = N_{const}/n_{local}$ is the same for different open locations, which makes the average molecular number in each cell (with larger or lower n) almost uniform (DSMC cannot do this and will have less simulated molecules in cells with smaller n due to $N_i(t) \equiv N_{init}$; but this DSBGK option is useful when two open ends have a large contrast of local n , in which DSMC will have large noise in the cells of small n and thus needs a large sampling size due to the unbalanced noise distribution).

Some techniques for enhanced performance:

- Slightly change the random number array (e.g., delete a few) if the density drift occurs in closed problems.
- Stop updating T_k for isothermal problems to significantly improve the robustness.
- Or apply smoothing technique in updating the noisy T_k for general problems (didn't try yet but should work).
- Using more simulated molecules per cell (instead of 10/cell) will always improve robustness.

